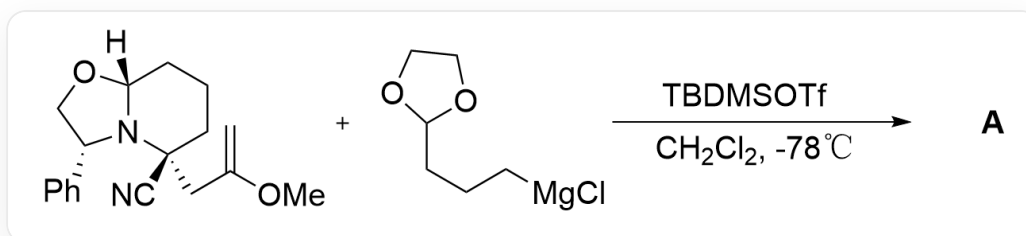


Question

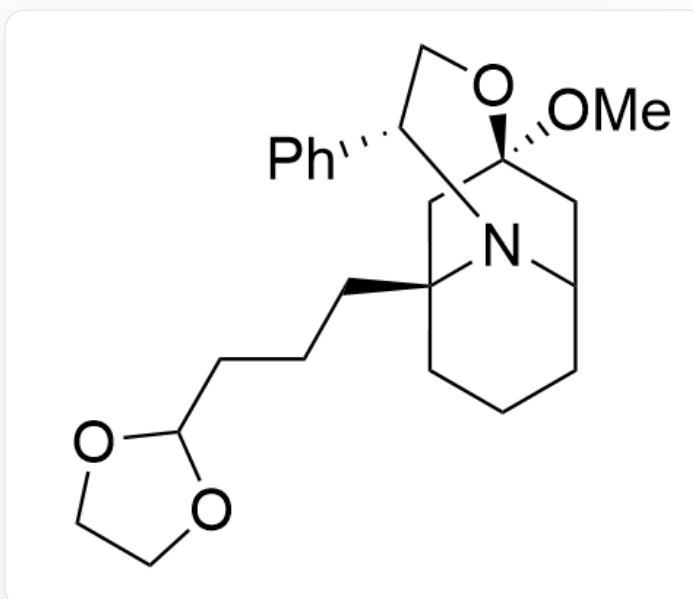


[H][C@]1(OC[C@H]2C3=CC=CC=C3)N2[C@](CC(OC)=C)(C#N)CCC1.Cl[Mg]CCCC4OCCO4>C[Si](C)(OS(=O)(C(F)(F)F)=O)C(C)(C)C.ClCCl>[**A**], **A** is the reaction product, reaction temperature -78°C

Given that the molecular formula of the reaction product **A** is C₂₃H₃₃NO₄, provide the structural formula of **A**.

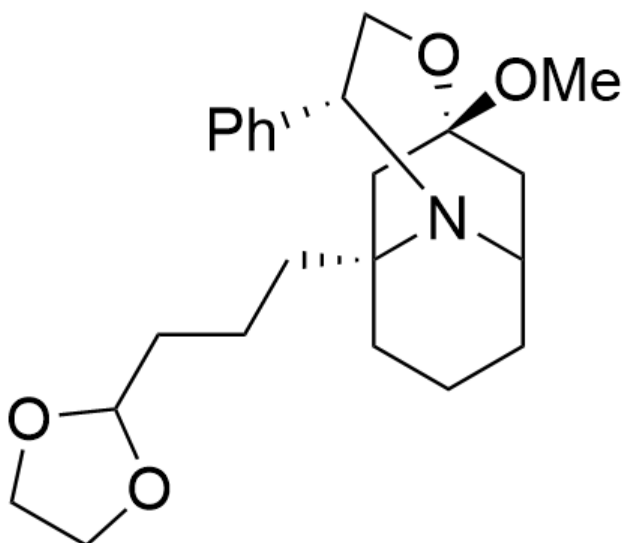
A. All other options are incorrect

B.



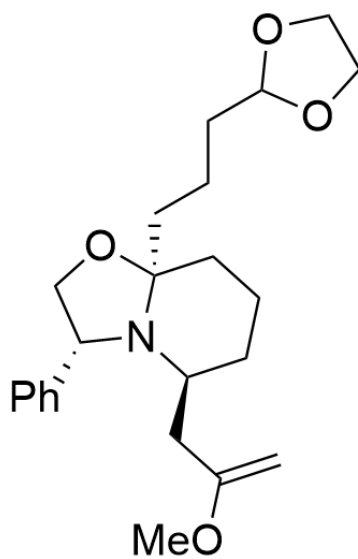
CO[C@]1(C2)C[C@@]3(CCCC4OCCO4)CCCC2N3[C@H](C5=CC=CC=C5)CO1

C.



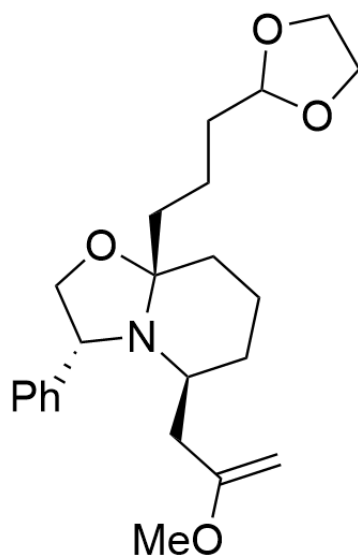
CO[C@@]1(C2)C[C@]3(CCCC4OCCO4)CCCC2N3[C@H](C5=CC=CC=C5)CO1

D.



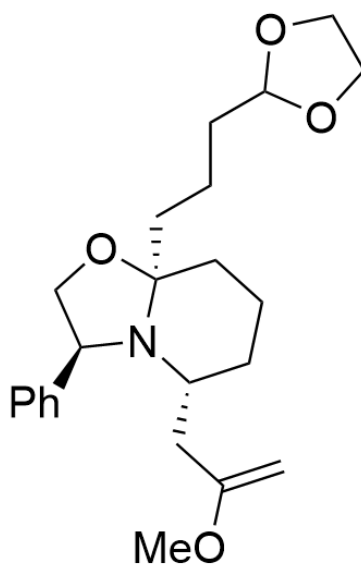
C=C(OC)C[C@H]1N2[C@](OC[C@H]2C3=CC=CC=C3)(CCCC4OCCO4)CCC1

E.



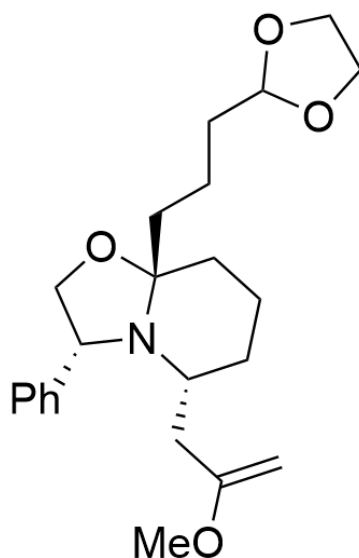
C=C(OC)C[C@H]1N2[C@@](OC[C@H]2C3=CC=CC=C3)(CCCC4OCCO4)CCC1

F.



C=C(OC)C[C@@H]1N2[C@](OC[C@@H]2C3=CC=CC=C3)(CCCC4OCCO4)CCC1

G.



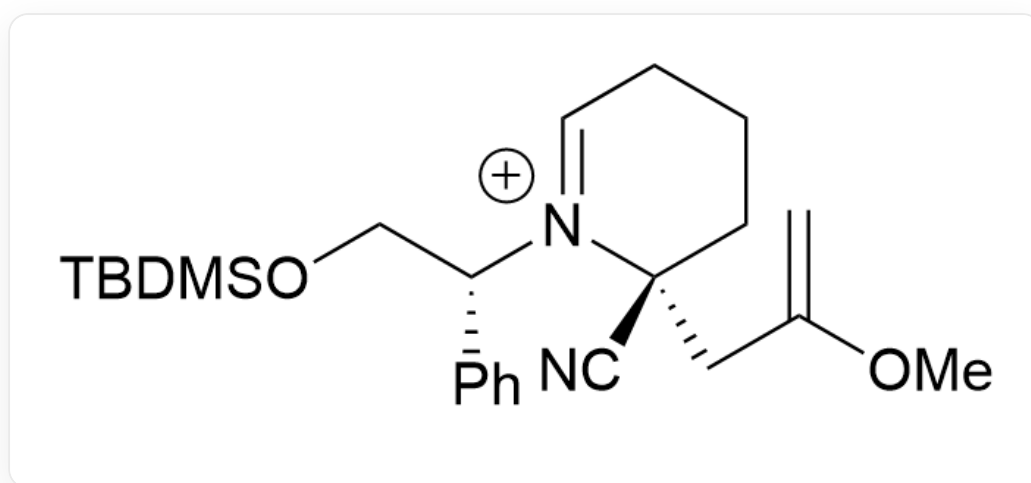
C=C(OC)C[C@@H]1N2[C@@](OC[C@H]2C3=CC=CC=C3)(CCCC4OCCO4)CCC1

Answer

Correct Answer: **D**

Detailed Explanation

First, the substrate is activated by one molecule of TBDMSOTf to obtain intermediate **1**



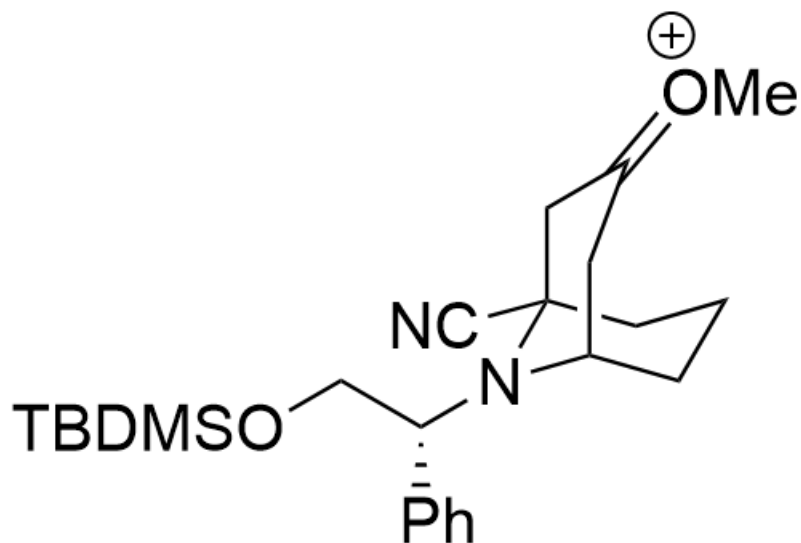
intermediate1 : C=C(OC)C[C@]1(C#N)[N+](C(=C)C=C)C(C)C(C)C(C)C=C1

CHECKPOINT

1 PTS

intermediate **1** : C=C(OC)C[C@]1(C#N)[N+](C(=C)C=C)C(C)C(C)C(C)C=C1

Then, the carbocation is captured by the adjacent alkenyl group to obtain intermediate **2**



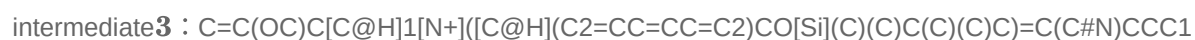
intermediate **2** : C[O+]=C1C[C@]2(C#N)N([C@H](C3=CC=CC=C3)CO[Si](C)(C)C(C)(C)C)[C@H](C1)CCC2

CHECKPOINT

1 PTS

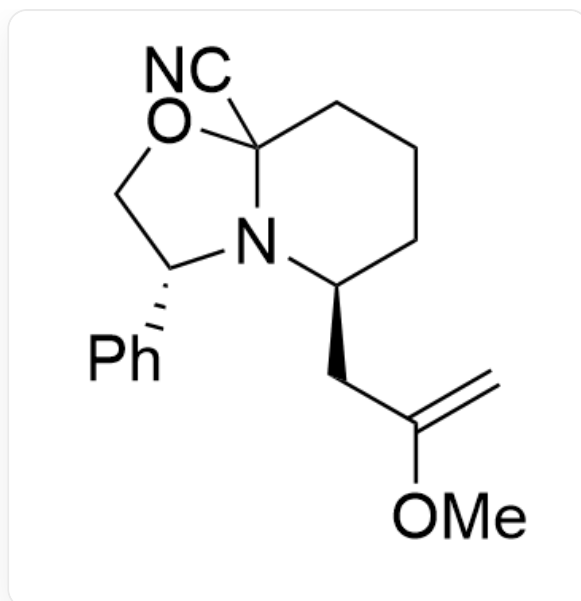
intermediate **2** : C[O+]=C1C[C@]2(C#N)N([C@H](C3=CC=CC=C3)CO[Si](C)(C)C(C)(C)C)[C@H](C1)CCC2

Then, ring opening occurs from the other direction to obtain intermediate **3**



1 PTS

Then the protecting group on the oxygen is removed and cyclization occurs to obtain intermediate **4**



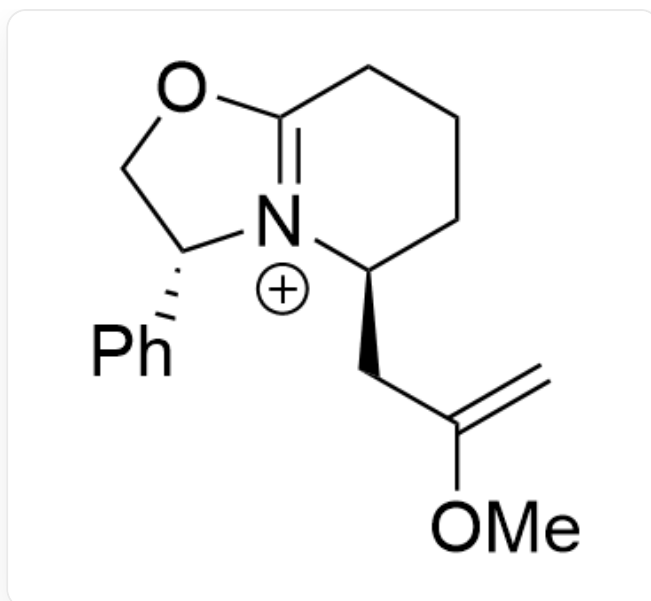
intermediate4 : C=C(OC)C[C@H]1N2C(OC[C@H]2C3=CC=CC=C3)(C#N)CCC1

CHECKPOINT

1 PTS

intermediate 4 : C=C(OC)C[C@H]1N2C(OC[C@H]2C3=CC=CC=C3)(C#N)CCC1

This intermediate can lose a cyanide ion to obtain intermediate 5



intermediate5 : C=C(OC)C[C@H]1[N+]2=C(OC[C@H]2C3=CC=CC=C3)CCC1

The last step of Grignard reagent addition is irreversible

CHECKPOINT

1 PTS

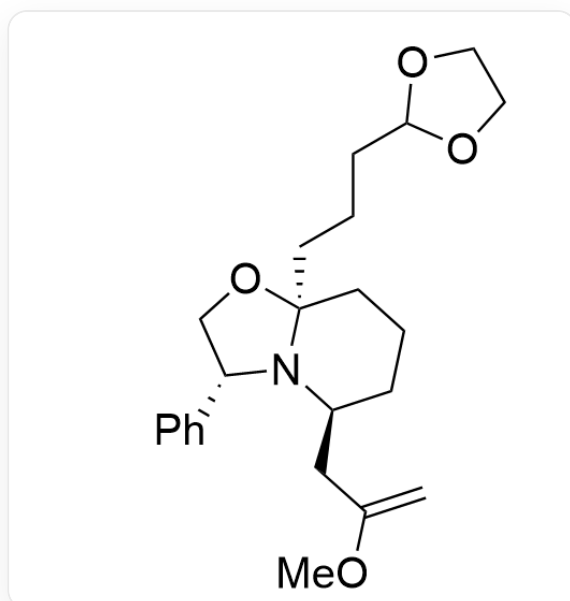
The last step of Grignard reagent addition is irreversible

Since the side containing the alkyl chain has larger steric hindrance, and the complexation with Mg^{2+} may further increase the steric hindrance on the side of the alkyl chain, the Grignard reagent tends to attack from the side opposite to the alkyl chain to obtain the reaction product **A**

CHECKPOINT

1 PTS

Since the side containing the alkyl chain has larger steric hindrance, and the complexation with Mg^{2+} may further increase the steric hindrance on the side of the alkyl chain, the Grignard reagent tends to attack from the side opposite to the alkyl chain to obtain the reaction product **A**



product **A** : C=C(OC)C[C@H]1N2[C@](OC[C@H]2C3=CC=CC=C3)(CCCC4OCCO4)CCC1